



Bruno Ferreira

Applied AI Engineer / AI Solutions Engineer

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PROFILE Applied AI Engineer focused on NLP, language models and document automation. I first learn how the work is done, then build a practical solution and test it with the people who will use it. My background combines research, software engineering, teaching and client work. I am primarily targeting roles in Portugal, while remaining open to opportunities elsewhere in the European Union.

Applied AI | NLP and language models | Document automation | AI evaluation | Full-stack delivery

01 / PROFESSIONAL EXPERIENCE

Independent Applied AI Consultant and Trainer

2025 - Present

Selected client work | Portugal and remote

- Design and build applied AI solutions for document-heavy and knowledge-intensive work, from initial discovery and prototyping to validation and deployment preparation.
- Deliver practical AI training and turn operational needs into clear requirements, evaluation criteria and points for human review.

Invited Lecturer

February 2024 - Present

University of Coimbra

- Modules taught: Principles of Databases, Fundamentals of Artificial Intelligence, Principles of Java, Principles of C, Natural Language Processing and Knowledge Representation.
- Preparation of learning materials, guidance and assessment of student work, and supervision of MSc dissertations.

Researcher

Feb 2022 - Jun 2025

CISUC | Centre for Responsible AI

- Conducted research in NLP, NLG, Transformers, Information Extraction, Question Answering, language models, Explainable AI, causal inference and model compression.
- Published and presented work on Question Answering, information extraction and the evaluation of knowledge encoded in language models.

02 / SELECTED APPLIED AI PROJECTS

Privacy-by-design AI platform

Product in development

CONTEXT: Own product combining local personal-data detection and anonymisation with external language-model workflows.

CONTRIBUTION: Defined the product strategy and architecture and built a functional PWA with document processing, model-assisted chat, projects, user profiles and verifiable privacy controls.

TOOLS: React, TypeScript, NER, ONNX, LLMs, Privacy by Design

Quality Copilot for document validation

2026

CONTEXT: Specialised modules that support comparison and validation of packaging and technical documentation. **CONTRIBUTION:** Built workflows for dimensional checks, content validation, correction verification and technical-sheet preparation, with human review at every stage.

TOOLS: Python, OCR, Claude Code skills, document automation

AI platform for US visa workflows

2025 - Present

CONTEXT: Multi-tenant LegalTech platform that applies USCIS criteria to scattered documentation in a consistent, evidence-guided workflow. **CONTRIBUTION:** Worked on the design and implementation of intake, triage, questionnaires, document management, OCR, language-model analysis and reports for legal review.

TOOLS: FastAPI, Next.js, PostgreSQL, OCR, LLMs

03 / SELECTED APPLIED AI PROJECTS | PART 2

Global THE 17.3 evidence evaluation

2025

CONTEXT: Commissioned to design and carry out a systematic review of public university information for THE Impact Rankings indicators 17.3.1 to 17.3.17. **CONTRIBUTION:** Defined the evidence-review method and a repeatable assessment process, applied consistently across the 17 indicators.

TOOLS: Evidence review, evaluation design, structured analysis

04 / TECHNICAL CAPABILITIES

Applied AI: NLP, language models, Information Extraction, Question Answering, OCR, XAI, prompting, evaluation and human review

Engineering: Python, TypeScript, React, Next.js, FastAPI, SQL, PostgreSQL, APIs, Git, Unix, Docker and document automation

From problem to delivery: Needs discovery, prototyping, testing, technical review, documentation, deployment preparation and training

Development with AI tools: Claude Code, Codex, OpenCode and Pi for implementation, review, testing and documentation, with defined review criteria and my own supervision

05 / EDUCATION

PhD in Informatics Engineering

2022 - Present

University of Coimbra

MSc in Informatics Engineering - Intelligent Systems

2022

University of Coimbra

BSc in Informatics Engineering

2021

University of Coimbra

06 / SELECTED PUBLICATIONS

- Towards Automated Evaluation of Knowledge Encoded in Large Language Models | LREC-COLING, 2024
- BATS-PT: Assessing Portuguese Masked Language Models | PROPOR, 2024
- Leveraging Question Answering for Domain-Agnostic Information Extraction | CIARP, 2023
- Evaluating the Extraction of Toxicological Properties with Extractive Question Answering | EANN, 2023
- SafetyDesk: Accelerating the Production of Toxicological Reports | EXP.AT, 2023
- ETFs Information Extraction from KIIDs | Environmental Finance for the Common Good, 2023
- Question Answering for Toxicological Information Extraction | SLATE, 2022
- Safety Desk: Extraction and Analysis of Textual Information to Build a Reporting System | MSc dissertation, 2022

07 / LANGUAGES

Portuguese - Native | English - Advanced | Spanish - Good comprehension, basic speaking

08 / PROFESSIONAL LINKS

[Portfolio](#) | [Ciência Vitae](#) | [ORCID](#)
